

Deepak Sathyan

Publication List

Articles in Review

5. **Intensity-Frontier Signals of Warped Extra Dimensions** Aug 2026
D. Kim, D. Sathyan, A. Verma
arxiv.org/abs/2608.20482
4. **Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering** May 2026
B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan
arxiv.org/abs/2605.08010
3. **A Baryon and Lepton Number Violation Model Testable at the LHC** Aug 2025
A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
arxiv.org/abs/2508.21064
2. **New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis** July 2025
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2507.01000
1. **United States Muon Collider Community White Paper for the European Strategy for Particle Physics Update** Mar 2025
M. Begel et al.
arxiv.org/abs/2503.23695

Articles in Refereed Journals

5. **New laboratory constraints on neutrinophilic mediators** July 2024
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)
4. **‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W** Apr 2024
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)
3. **A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET** Oct 2023
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)
2. **Energy-peak based method to measure top quark mass via B-hadron decay lengths** Dec 2022
K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)

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| 1. LHC Signals for KK Graviton from an Extended Warped Extra Dimension | Aug 2020 |
| K. Agashe, M. Ekhaterachian, D. Kim, D. Sathyan | |
| arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109) | |

Articles in Preparation

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| 1. LHC as a Beam Dump: Probing Light Dark Photons | Sept 2026 |
| B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan | |
| arxiv.org/a/sathyan_d_1 | |
| 2. New Probes of Dark Sectors at a Muon Collider | Jan 2027 |
| B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan | |
| arxiv.org/a/sathyan_d_1 | |
| 3. Exploring open systems approach to neutrino oscillations | Mar 2027 |
| A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou | |
| arxiv.org/a/sathyan_d_1 | |
| 4. LHC Signals for KK Gravitons Producing 8-Particle Final States | Mar 2027 |
| K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan | |
| arxiv.org/a/sathyan_d_1 | |

Ph.D. Thesis

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| 1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass | 2024 |
| D. Sathyan | |
| drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis) | |

Snowmass2021 Contributions

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| 5. TF07 Snowmass Report: Theory of Collider Phenomena | Oct 2022 |
| F. Maltoni et al. | |
| arxiv.org/abs/2210.02591 | |
| 4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021 | Sept 2022 |
| K. Agashe et al. | |
| arxiv.org/abs/2209.11267 | |
| 3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021 | Sept 2022 |
| T. Bose et al. | |
| arxiv.org/abs/2209.13128 | |
| 2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond | Apr 2022 |
| K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan | |
| arxiv.org/abs/2204.02928 | |

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| 1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models | Mar 2022 |
| K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan | |
| arxiv.org/abs/2203.13305 | |

Muon g-2 Articles

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| 3. The straw tracking detector for the Fermilab Muon g-2 Experiment | Nov 2021 |
| B.T. King et al. | |
| arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035) | |
| 2. Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab | Apr 2021 |
| Muon g-2 Collaboration | |
| arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002) | |
| 1. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm | Apr 2021 |
| Muon g-2 Collaboration | |
| arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801) | |